• On execution of Case 4, there are $(x+1)$ even, $(y-2)$ odd integers.

$\therefore$ Even integers $\longrightarrow$ parity changed.

Odd integers $\longrightarrow$ same parity.

$\therefore$ After each step in the process, the parity of even integers change but the parity of odd integers remain same.

$\because$ Initially, there are n even, n odd integers and n is odd.

Finally, after $(2n-1)$ steps there is only 1 integer. $\because$ 1 is an odd number, the final integer must be odd. □

28. Suppose that five ones and four zeros are arranged around a circle. Between any two equal bits you insert a 0 and between any two unequal bits you insert a 1 to produce nine new bits. Then you erase the nine original bits. Show that when you iterate this procedure, you can never get nine zeros. [Hint: Work backward, assuming that you did end up with nine zeros.]

Solution: Let, we have 9 positions around a circle namely P1, P2, P3, P4, P5, P6, P7, P8 and P9. Suppose, we finally get 9 0's as shown below

0	0	0	0	0	0	0	0	0
P1	P2	P3	P4	P5	P6	P7	P8	P9

$\therefore$ In the previous step, by defn, all nine positions could contain 0 because between any two 0's, there would be a 0.

Let, Previous P1 = 1 $\therefore$ By given condition, Previous P2 = 1.

By the same condition, Prev P3 = Prev P4 = Prev P5 = Prev P6 = Prev P7 = Prev P8 = Prev P9 = 1.

$\therefore$ In the previous step, the arrangement is shown below:

1	1	1	1	1	1	1	1	1
P1	P2	P3	P4	P5	P6	P7	P8	P9